

A no-go for perturbative baryogenesis in cosmologies with a bounded thermal history

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Abstract

A class of cosmological models — bounce scenarios with a density cap near the QCD scale, cyclic models, and low-reheating histories — never reaches the temperatures at which the standard baryogenesis mechanisms operate. We quantify what remains. For any thermal history bounded by $T_{\max} \lesssim 130$ GeV the Standard-Model sphalerons, thermal leptogenesis and GUT-scale mechanisms are unavailable in principle, so the asymmetry must be produced by baryon-number-violating interactions active at $T \leq T_{\max}$. For the representative dimension-six operator $(qqq\ell)/\Lambda^2$, cosmological relevance at $T_{\max} \sim \Lambda_{\text{QCD}}$ requires $\Lambda \lesssim 3 \times 10^6$ GeV even under a maximal thermal chemical-potential bias (efficiency floor $\epsilon_{\min} = 1.4 \times 10^{-9}$), while proton stability, $\tau_{p \rightarrow e^+ \pi^0} > 2.4 \times 10^{34}$ yr, requires $\Lambda \gtrsim 3 \times 10^{16}$ GeV: the two windows are separated by ten orders of magnitude in Λ , i.e. forty orders in τ_p . We show that environmental “gating” of the operator cannot bridge the gap — power-law gates tied to the chiral condensate fall short by ~ 20 orders of magnitude and switch *on* inside nucleons in bag-like descriptions, while derivative (phase-velocity) gates are radiatively unstable — and we classify the surviving mechanisms exhaustively: (i) B – L -conserving transfer of the asymmetry to a dark sector with kinematic nucleon protection, (ii) condensate-carried baryon number of the Affleck–Dine type, and (iii) non-thermal populations with $E \gg T_{\max}$. Published bounce-baryogenesis scenarios that assume equilibrium B -violation below the sphaleron threshold are excluded unless recast in one of these classes. In cyclic bounded- T models a primordial asymmetry is additionally excluded by entropy dilution, forcing per-cycle regeneration. As a practical consequence, the mere coexistence of a baryon asymmetry with a stable proton becomes a thermometer of the early universe: it implies either an epoch above the sphaleron threshold or the existence of one of the three structures above.

1 Introduction

Any dynamical explanation of the baryon asymmetry of the universe, $\eta_B = n_B/n_\gamma = (6.12 \pm 0.04) \times 10^{-10}$ [1], must satisfy the three Sakharov conditions [2]. The third condition — baryon-number violation in action — is supplied in the standard scenarios either by the electroweak sphalerons, which freeze out at $T_{\text{sph}} \simeq 132$ GeV [3], or by heavy states at the grand-unification scale. Both options presuppose that the universe was, at some epoch, hot enough.

A growing class of cosmological models does not grant that presupposition. Non-singular bounce models in which the energy density is capped at a microphysical scale — in particular QCD-scale bounces with $\rho_c \sim (1 \text{ GeV})^4$, for which the maximal temperature is $T_{\max} = (30\rho_c/\pi^2 g_*)^{1/4} \sim 400$ MeV — never pass through an electroweak or GUT epoch,

in this aeon or, for cyclic versions, in any aeon. Low-reheating scenarios present the same situation for a different reason: if the inflaton or a late modulus reheats the universe only to $T_{\text{RH}} \ll 100$ GeV, the standard mechanisms are equally out of reach; this regime has been studied since the work of Ref. [4] and remains viable down to $T_{\text{RH}} \simeq 5$ MeV.

Faced with a bounded thermal history, model builders have frequently appealed to *gravitational* or *spontaneous* baryogenesis operating near the bounce [5, 6], following the mechanism of Ref. [7]: a time-dependent background (curvature or a rolling phase) supplies an effective chemical potential for baryon number, and a B -violating interaction in equilibrium converts it into an asymmetry that freezes out at a decoupling temperature T_D , treated as a free parameter. The purpose of this paper is to point out, quantitatively, that the second ingredient is not available for free: *any* perturbative B -violating interaction fast enough to matter at $T_D \leq T_{\text{max}} \lesssim 130$ GeV makes the proton decay at a catastrophic rate. The qualitative tension between low-scale B -violation and proton stability is old — it was decisive, for instance, for baryogenesis in low quantum-gravity-scale models [8] — but to our knowledge it has not been formulated as an exclusion map for bounded thermal histories, nor confronted with the gating strategies that a bounce environment naturally suggests, nor combined with the entropy-dilution argument that closes the initial-condition escape in cyclic models.

The paper is organized as follows. Section 2 defines bounded thermal histories and disposes of the Standard-Model channels. Section 3 derives the exclusion map in the $(T_{\text{max}}, \Lambda)$ plane (Fig. 1), whose two allowed regions never intersect below the sphaleron threshold. Section 4 shows that environment-dependent operator coefficients cannot rescue the mechanism. Section 5 classifies the surviving routes and gives a worked benchmark of the dark-transfer class, including its laboratory signature. Section 6 proves the corollary for cyclic models. Section 7 collects the practical implications, and Section 8 concludes. All numerical statements are reproducible from the public repository accompanying this paper.

2 Bounded thermal histories

Definition. A cosmology has a *bounded thermal history* if the temperature of the radiation bath satisfies $T(t) \leq T_{\text{max}}$ for all times after the initial (or bounce) epoch, with $T_{\text{max}} < T_{\text{sph}} \simeq 132$ GeV, while remaining above the BBN floor $T_{\text{max}} \gtrsim 5$ MeV so that standard nucleosynthesis proceeds.

For a bounce whose density is capped at ρ_c the bound is structural: with $g_* \simeq 47.5$ light degrees of freedom at the QCD scale, $\rho_c = (0.86 \text{ GeV})^4$ gives $T_{\text{max}} = 432$ MeV, and in a cyclic realization the same cap applies to every aeon. The Standard-Model channels then close by exponentially large margins:

- *Electroweak sphalerons.* The vacuum transition rate is suppressed by $e^{-E_{\text{sph}}/T}$ with $E_{\text{sph}} \simeq 9$ TeV; at $T = 432$ MeV the exponent is $\simeq -2.1 \times 10^4$.
- *Electroweak instantons* coupled to any rolling phase (which would violate $B + L$): the vacuum rate carries $e^{-4\pi/\alpha_W} \simeq 10^{-164}$.
- *Thermal leptogenesis and GUT baryogenesis* require the thermal production of states with $M \gtrsim 10^9\text{--}10^{16}$ GeV and are unavailable by construction. (We note in passing that the QCD anomaly, to which a rolling strong-sector phase does couple, conserves baryon number: $SU(3)_c$ is vector-like.)

Baryogenesis in a bounded history therefore requires an interaction violating B (or transferring it) that is active at $T \leq T_{\max}$. We now show that the perturbative version of this requirement is self-destructive.

3 The EFT window versus proton stability

The lowest-dimension baryon-number-violating operators built from Standard-Model fields are the dimension-six four-fermion structures $\mathcal{O}_6 = (qqq\ell)/\Lambda^2$. In a bath at temperature T they mediate $\Delta B \neq 0$ scatterings at a rate $\Gamma_B \sim T^5/\Lambda^4$.

Cosmological relevance. The most efficient use of such an operator in a bounded history is spontaneous baryogenesis [9]: a time-dependent background provides an effective chemical potential μ_B , and the operator converts it into an asymmetry $n_B/s \simeq (\mu_B T^2/6)/s \cdot \min(1, \Gamma_B/H)$. The bias cannot exceed its thermal ceiling $\mu_B \lesssim \pi T$ (beyond which the linear-response expression saturates), which gives the *efficiency floor*

$$\epsilon_{\min} = \frac{\eta_{\text{obs}}^{(s)}}{\eta_{\text{eq}}^{\max}} = \frac{8.6 \times 10^{-11}}{45/(12\pi g_*)} \simeq 1.4 \times 10^{-9}, \quad g_* = 20, \quad (1)$$

where $\eta^{(s)} \equiv n_B/s$. Demanding $\Gamma_B/H \geq \epsilon_{\min}$ at $T_{\max}$ bounds the operator scale from above:

$$\Lambda \leq \Lambda_{\text{viab}}(T_{\max}) = \left(\frac{T_{\max}^3 M_{\text{Pl}}}{1.66\sqrt{g_*}} \right)^{1/4} \epsilon_{\min}^{-1/4} = 3.1 \times 10^6 \text{ GeV} \quad \text{at } T_{\max} = 432 \text{ MeV}. \quad (2)$$

Proton stability. The same operator mediates $p \rightarrow e^+\pi^0$ with $\tau_p \sim \Lambda^4/m_p^5$. The Super-Kamiokande bound $\tau_{p \rightarrow e^+\pi^0} > 2.4 \times 10^{34} \text{ yr}$ [10] requires

$$\Lambda \geq \Lambda_p = (\tau_{\text{SK}} m_p^5)^{1/4} = 3.0 \times 10^{16} \text{ GeV}. \quad (3)$$

Theorem 1. *In a cosmology with $5 \text{ MeV} \lesssim T_{\max} < T_{\text{sph}}$, no dimension-six baryon-number-violating operator of Standard-Model fields can be simultaneously cosmologically relevant, Eq. (2), and consistent with proton stability, Eq. (3). At $T_{\max} = 432 \text{ MeV}$ the two windows are separated by a factor 10^{10} in Λ , i.e. forty orders of magnitude in τ_p .*

Figure 1 shows the exclusion map. Two features deserve comment. First, the conclusion is insensitive to the order-of-magnitude character of the rate estimates: a generous 10^3 in Γ_B moves Λ_{viab} by a factor 5.6 against a gap of 10^{10} . Higher-dimension operators only widen the gap. Second, formally the “EFT desert” closes only at $T \sim 9 \times 10^{12} \text{ GeV}$; long before that, at $T > T_{\text{sph}}$, the Standard Model supplies its own non-perturbative B -violation, whose exponential activation is precisely the kind of vacuum-safe behaviour a perturbative operator cannot imitate (Sec. 4). The theorem thus *recovers* the familiar lore — viable baryogenesis lives either at the sphalerons or at the GUT scale — and identifies its reason below the electroweak scale: the intermediate range is proton-forbidden.

4 Gating does not rescue the operator

A bounce environment suggests an escape: let the operator coefficient be large at the bounce and small in vacuum, $\mathcal{O}_6 \rightarrow G(x) \mathcal{O}_6$, with a gate G built from the ambient fields.

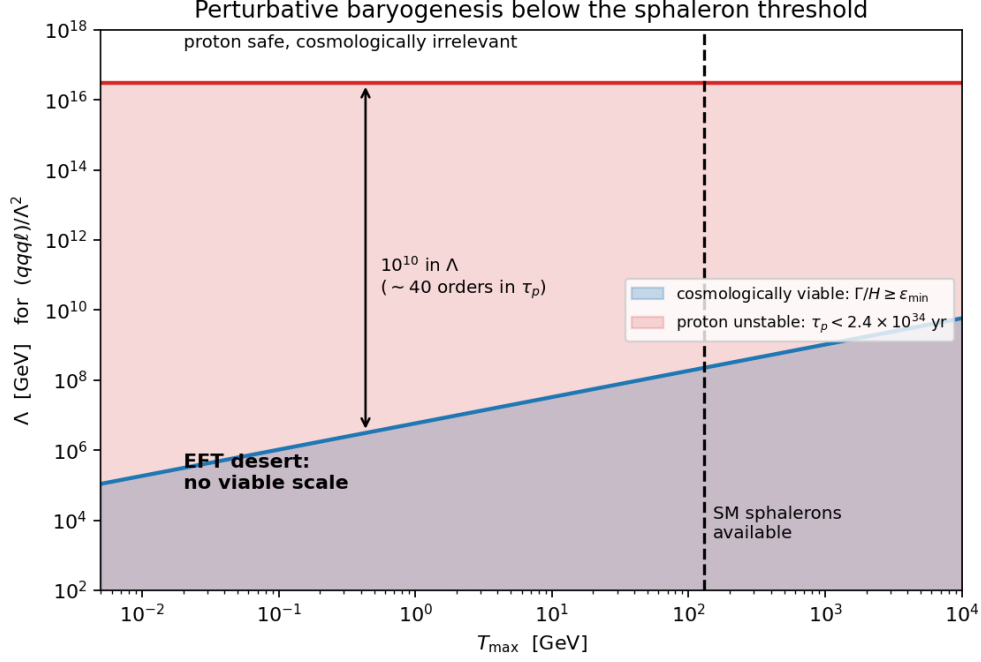


Figure 1. The $(T_{\max}, \Lambda)$ plane for the operator $(qqq\ell)/\Lambda^2$. Blue region: cosmologically capable of generating the observed asymmetry under a maximal thermal bias, Eq. (2). Red region: proton lifetime below the Super-Kamiokande bound, Eq. (3). The regions never meet below the sphaleron threshold (dashed line); at $T_{\max} = 432$ MeV the gap is ten orders of magnitude in Λ .

Closing a 10^{10} gap in Λ requires the vacuum amplitude suppressed by $G_{\text{vac}} \lesssim 10^{-22}$ relative to the bounce. Both natural gate families fail.

Environmental (condensate) gates. Take $G = (1 - \sigma^2/\sigma_0^2)^n$ with σ the chiral condensate, so that $G \rightarrow 1$ where the condensate melts (the bounce) and $G \rightarrow 0$ in vacuum. The flaw is that the proton’s interior is not vacuum: in-medium condensate reduction at nuclear density is 20–30 %, so the gate base inside a nucleon is $G^{1/n} \simeq 0.25\text{--}0.3$, and $G \leq 10^{-22}$ requires $n \gtrsim 37$ powers of the field — outside any effective field theory. Worse, in bag-model descriptions the condensate is strongly suppressed *inside* hadrons, turning the gate fully on precisely where it must vanish.

Derivative gates are radiatively unstable. Take $G = (\partial\theta)^2/M^4$ with θ a phase that rolls at the bounce and is static today; tree-level proton decay then vanishes in vacuum. But no symmetry distinguishes the gated from the ungated operator (both are invariant under the shift symmetry of θ), so vacuum fluctuations regenerate the ungated coefficient at one loop: $\langle(\partial\theta)^2\rangle \sim f^2/16\pi^2$ with f the decay constant, exceeding any natural normalization $M^4 \sim m_\theta^2 f^2$ by two orders of magnitude for QCD-scale parameters. The gate is a fine-tuning, not a mechanism. (The sphaleron’s own exponential activation evades this argument because it is a property of the gauge dynamics, not an operator coefficient.)

5 The surviving mechanisms

The theorem constrains *perturbative, thermal* B -violation. Exactly three structures evade its assumptions.

5.1 Class I: $B-L$ -conserving dark transfer

Instead of violating baryon number, transfer it. Introduce a dark fermion χ carrying baryon (or $B-L$) charge and the neutron portal

$$\mathcal{O}_{\text{tr}} = \frac{(udd)\chi}{\Lambda_{\text{tr}}^2}, \quad (4)$$

which conserves the *total* charge — so no proton-decay operator is generated at any order — while violating the visible–dark partition that a background bias can act on. Spontaneous *cogenesis* then separates a net zero into equal and opposite asymmetries, $N_B = N_\chi$. Nucleon and χ stability are guaranteed kinematically if the dark-fermion mass lies in the corridor

$$m_n - S_n(^9\text{Be}) < m_\chi < m_p + m_e \implies 937.900 < m_\chi < 938.783 \text{ MeV}, \quad (5)$$

the window familiar from the neutron dark-decay literature [11]: bound neutrons in the most fragile nucleus cannot convert, and $\chi \rightarrow pe\bar{\nu}$ is closed. A worked benchmark with the bias supplied by the QCD phase (the η' direction) at a QCD-scale bounce gives $\Lambda_{\text{tr}} \simeq 1\text{--}2 \times 10^3 \text{ TeV}$ and, unavoidably, a mixing-induced free-neutron channel

$$\text{Br}(n \rightarrow \chi\gamma) \sim 10^{-8}\text{--}10^{-6}, \quad (6)$$

a genuine laboratory floor: lowering Λ_{tr} to strengthen cogenesis raises the branching as Λ_{tr}^{-4} . This class is not hypothetical — B -mesogenesis [12, 13] realizes the same charge bookkeeping with B mesons and is actively searched for at colliders; asymmetric dark matter [14] is its relic-abundance-oriented relative.

5.2 Class II: condensate-carried baryon number

The Affleck–Dine mechanism [15] evades the theorem because the asymmetry is generated by the coherent dynamics of a B -charged scalar, not by thermal scatterings: the “operator” acts on a macroscopic field displacement, and the transfer to quarks proceeds through renormalizable couplings. In a bounded history the usual challenge is inverted: without a high-temperature epoch, the mechanism must explain how the initial displacement is set — in a bounce, e.g. by the contraction phase — and why the A -terms do not induce low-energy B -violation elsewhere. We do not pursue this here beyond noting that it remains logically open and requires a dedicated field content.

5.3 Class III: non-thermal high-energy populations

The bath may be cold while individual quanta are not. Three known realizations: late decays of heavy relics (moduli reheating to $T_{\text{RH}} \sim 200 \text{ MeV}$ with baryogenesis from the decays themselves [16]); collision-induced electroweak sphalerons, where a non-thermal particle with $E_{\text{cm}} > E_{\text{sph}}$ scatters off the bath [17]; and evaporating primordial black holes. All share one requirement: a population with $E \gg T_{\text{max}}$ must be justified within the bounded history itself. In one-shot low-reheating cosmologies the inflaton or modulus supplies it naturally; in cyclic bounded models it must be *regenerated* every aeon — see below.

6 Corollary for cyclic cosmologies

Cyclic models offer a tempting escape: take η_B as an initial condition inherited from before. Entropy production closes it.

Corollary 1. *In a cyclic cosmology whose comoving entropy grows by a factor $g > 1$ per aeon while comoving baryon number is conserved, the asymmetry dilutes as $\eta_n^{(s)} = \eta_1^{(s)} g^{-(n-1)}$. For $g = 2$ and $n = 77$ aeons, reproducing $\eta_{\text{obs}}^{(s)} = 8.6 \times 10^{-11}$ today requires $\eta_1^{(s)} \simeq 6 \times 10^{12}$ — more than one baryon per entropy quantum, which is impossible. Inheritance is excluded by thirteen orders of magnitude; the asymmetry must be regenerated every cycle through Class I or II (Class III populations dilute alike).*

The growth factor g is model-specific, but any $g \geq 1.2$ excludes inheritance over $\gtrsim 150$ aeons; entropy production is the defining feature of cyclic scenarios, so the corollary has wide reach.

The same dilution squeezes Class III in cyclic settings. The collision-sphaleron route of Ref. [17] requires quanta with $E_{\text{cm}} \gtrsim E_{\text{sph}} \simeq 9$ TeV, and the relic-decay route requires long-lived states with $M \gg T_{\text{max}}$; a bath capped at $T_{\text{max}} \sim 400$ MeV cannot produce either thermally, and any non-thermal population prepared in one aeon dilutes away like the asymmetry itself. A cyclic realization of Class III therefore needs a *per-cycle accelerator* — e.g. black-hole formation and evaporation during the crunch, or the bounce dynamics itself — whose energetics must be demonstrated, not assumed. Cyclic bounded- T cosmologies are thus pushed toward Classes I and II every aeon, with Class I offering the only generic laboratory handle.

7 Practical implications

A triage rule for model builders and referees. Any bounce, cyclic or low-reheating proposal that explains η_B must now declare which of the three surviving classes it realizes. A paper that posits a B -violating rate $\Gamma_B = T^{2m+1}/M_B^{2m}$ with a freely chosen decoupling temperature $T_D < T_{\text{sph}}$ — the standard parameterization of the gravitational-baryogenesis literature applied to bounces [5, 6] — places itself inside the excluded region of Fig. 1 unless the operator is of the Class-I transfer type. The figure is intended as a one-look check.

The baryon asymmetry as a thermometer. Two low-energy facts — $\eta_B \neq 0$ and $\tau_p > 2.4 \times 10^{34}$ yr — jointly imply a statement about the deep past: *either* the universe once exceeded the sphaleron threshold, *or* nature realizes one of the three structures of Sec. 5 (a baryon-charged dark sector, a B -charged condensate, or a non-thermal high-energy population). Bounded-history cosmologies are therefore not unconstrained exotica: they inherit a sharp, falsifiable homework assignment.

Experimental targets. The no-go channels attention toward the observables of the survivors: the free-neutron dark branching of Class I ($\text{Br}(n \rightarrow \chi\gamma) \gtrsim 10^{-8}$ at QCD-scale cogenesis — within two to three orders of current search sensitivity, and a *floor* rather than a ceiling), the B -meson signatures of mesogenesis already pursued at B -factories [13], and the flat-direction phenomenology of Affleck–Dine sectors. A future confirmed detection of any of these would convert the weakest link of bounded-history cosmologies into their anchor; continued nulls progressively close the class.

8 Conclusions

Cosmologies whose thermal history never exceeds the electroweak sphaleron threshold cannot generate the baryon asymmetry through perturbative baryon-number-violating interactions of the thermal bath: at $T_{\text{max}} \sim \Lambda_{\text{QCD}}$ the operator scale required for cosmological relevance and the scale required by proton stability are separated by ten orders of magnitude, environmental gating of the operator is either insufficient by twenty orders or radiatively unstable, and in cyclic realizations the initial-condition escape is closed by entropy dilution. What survives is a short, exhaustive list — dark-sector transfer with kinematic protection, Affleck–Dine condensates, non-thermal high-energy populations — each with its own observational handle. The baryon asymmetry, combined with proton stability, thereby acts as a thermometer of the maximum temperature of the universe: whoever removes the hot epoch must add one of these three structures, visibly.

Code and data availability

All numerical statements are reproducible from the public repository github.com/infosave2007/vmf: the exclusion map of Fig. 1 (`papers/bounded_T_nogo/make_exclusion_figure.py`), the channel-by-channel closure (`verification/nvg_baryogenesis_bsm_closure.py`), the Class-I benchmark (`verification/nvg_adm_bl_cogenesis.py`) and the cyclic dilution corollary (`verification/nvg_etab_inheritance.py`).

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